

HEARTBEAT MONITOR WITH ALERT SYSTEM

Embedded Systems Project Report

Project Title: Heartbeat Monitor with Alert System using ESP32

Project Type: Embedded Systems / Biomedical Embedded Systems Prototype

Development Platform: ESP32 DevKit V1 and Wokwi Simulation

Display: 0.96-inch I2C OLED

Alert System: Buzzer, Red LED and Green LED

1. ABSTRACT

- The "Heartbeat Monitor with Alert System" is an educational embedded systems project designed to demonstrate real-time heart-rate monitoring and threshold-based alert generation.
 - The system uses an ESP32 microcontroller as the main processing unit. A heartbeat-related input is processed by the microcontroller, the BPM (Beats Per Minute) value is determined, and the result is displayed on an OLED screen.
 - The system compares the BPM value with configurable educational thresholds. When the BPM is within the configured demonstration range, the system displays a NORMAL status and activates the green LED. When the BPM falls below the low threshold or rises above the high threshold, the system displays LOW or HIGH status, activates the red LED, and generates a buzzer alert.
 - Because an actual Pulse Sensor component is not available in the selected Wokwi simulation environment, the current virtual implementation uses software-generated BPM values to demonstrate normal, low and high heart-rate conditions.
 - This project demonstrates important embedded systems concepts including GPIO control, I2C communication, display interfacing, real-time processing, threshold comparison, alert generation, serial communication and simulation-based testing.
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2. INTRODUCTION

- Heart-rate monitoring is an important application area in healthcare, fitness, sports and wearable technology. Modern wearable devices continuously monitor heart-rate-related signals and provide users with information about their current physiological state.
 - The objective of this student project is to build a simplified embedded prototype that demonstrates the basic workflow of a heart-rate monitoring system.
 - The ESP32 acts as the central controller. The system receives or simulates heartbeat-related data, processes the BPM value, compares it with predefined educational thresholds and provides visual and audible feedback.
 - The project is designed as an educational prototype and demonstrates embedded-system design principles rather than providing clinical or medical functionality.
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3. PROBLEM STATEMENT

Continuous monitoring of heart rate can be useful in fitness tracking, wellness applications and healthcare-related monitoring systems.

However, a monitoring system must be able to:

- acquire heartbeat-related information,
- process the input,
- calculate or obtain BPM,
- determine whether the reading is within the configured range,
- display the result,
- and generate an alert when the reading crosses the configured limits.

The project addresses these requirements through an ESP32-based embedded monitoring prototype.

4. OBJECTIVES

The major objectives of this project are:

1. To design an ESP32-based heartbeat monitoring prototype.
2. To understand sensor-to-microcontroller interfacing.
3. To demonstrate BPM monitoring.

4. To display BPM information on an OLED.
 5. To implement configurable threshold logic.
 6. To activate a green LED during the normal demonstration condition.
 7. To activate a red LED during alert conditions.
 8. To activate a buzzer during alert conditions.
 9. To demonstrate the project through Wokwi simulation.
 10. To test normal, low and high BPM scenarios.
 11. To document the project as a GitHub proof-of-work project.
 12. To understand basic biomedical embedded-system concepts.
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5. EDUCATIONAL MEDICAL DISCLAIMER

- This project is an educational embedded-systems prototype.
 - It is NOT a medical diagnostic device, clinical monitoring device or certified healthcare product.
 - The BPM values used in the virtual simulation are generated for demonstration and testing purposes.
 - Hobby-grade sensors and educational prototypes may produce approximate readings and may be affected by sensor placement, movement, noise, lighting and other factors.
 - The configured LOW and HIGH thresholds in this project are demonstration values only. They must not be interpreted as medical advice or clinical diagnostic limits.
 - This prototype must not be used to make medical decisions.
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6. INDUSTRY RELEVANCE

Heartbeat monitoring concepts are used in many technology sectors.

6.1 Wearable Devices

Smartwatches and fitness bands monitor heart-rate-related signals and display information to users.

6.2 Fitness and Sports

Athletes and fitness applications can use heart-rate monitoring to track exercise intensity and training performance.

6.3 Healthcare Monitoring

Healthcare-related monitoring systems can use physiological sensors to collect information for professional assessment.

6.4 Rehabilitation

Monitoring technologies can support rehabilitation and wellness programs by providing physiological measurements.

6.5 Remote Monitoring

Connected embedded systems can transmit sensor information to remote platforms for monitoring and analysis.

6.6 Technical Industry Skills

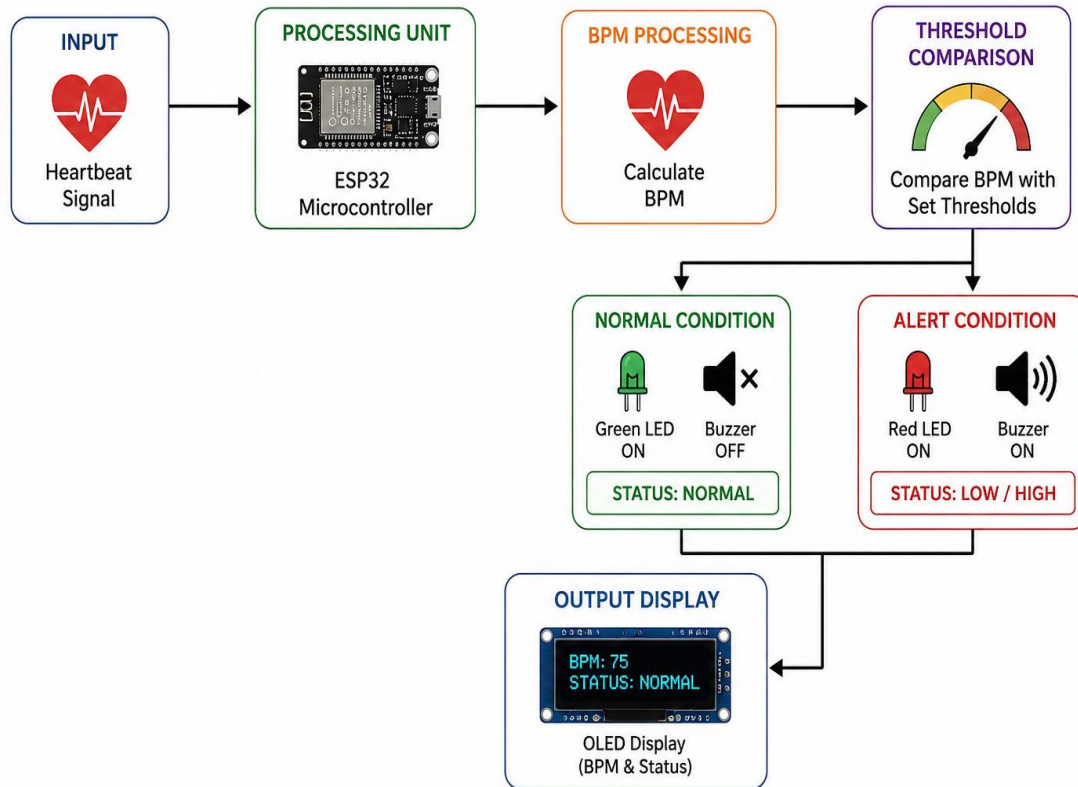
This project demonstrates skills relevant to embedded engineering:

- Microcontroller programming
 - GPIO control
 - Analog/digital signal concepts
 - I2C communication
 - Sensor interfacing
 - Real-time processing
 - Threshold-based decision logic
 - Display interfacing
 - Alert generation
 - Serial communication
 - Simulation
 - Testing and documentation
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7. SYSTEM OVERVIEW

SYSTEM OVERVIEW

The overall system follows this workflow:



8. HARDWARE COMPONENTS

8.1 ESP32 DevKit V1

Purpose

The ESP32 is the main microcontroller of the project.

Responsibilities

- Processing BPM data
- Controlling LEDs
- Controlling buzzer
- Communicating with OLED
- Running the alert logic
- Managing simulation logic

- Sending information to the Serial Monitor
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8.2 Pulse Sensor

Purpose

The Pulse Sensor is intended to detect pulse-related signals from the user.

Role in the final hardware concept

The sensor would provide an input signal to the ESP32, which would then process the signal to determine heartbeat timing and BPM.

Simulation limitation

The selected Wokwi environment does not provide the actual Pulse Sensor component used in the planned hardware implementation.

Therefore, the current Wokwi version uses software-generated BPM values to demonstrate the monitoring and alert logic.

8.3 OLED Display

Purpose

The OLED provides real-time visual information.

Example output

HEART MONITOR

BPM: 75

STATUS: NORMAL

During an alert:

HEART MONITOR

BPM: 110

STATUS: HIGH

8.4 Buzzer

Purpose

The buzzer provides an audible alert.

Behavior

Normal condition:

Buzzer = OFF

Low or high alert:

Buzzer = ON/OFF periodically

8.5 Green LED

The green LED indicates the configured normal condition.

NORMAL

|

v

Green LED ON

8.6 Red LED

The red LED indicates an alert condition.

LOW / HIGH

|

v

Red LED ON

8.7 220Ω Resistors

Two 220Ω resistors are used with the LEDs to limit current and protect the LEDs and GPIO outputs.

9. SOFTWARE REQUIREMENTS

The project uses:

- Arduino IDE / Arduino-compatible development environment
 - Wokwi simulation
 - Embedded C/C++ style Arduino programming
 - Adafruit SSD1306 library
 - Adafruit GFX library
 - Wire library
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10. LIBRARIES USED

10.1 Wire Library

The Wire library provides I2C communication functionality.

It is used to communicate between the ESP32 and OLED display.

10.2 Adafruit GFX Library

This library provides graphics and text drawing functionality used by the OLED display.

10.3 Adafruit SSD1306 Library

This library provides control functionality for SSD1306-based OLED displays.

11. PIN CONFIGURATION

The current Wokwi simulation uses the following configuration:

Component	ESP32 Pin
OLED SDA	GPIO 21
OLED SCL	GPIO 22
Buzzer	GPIO 25
Green LED	GPIO 26
Red LED	GPIO 27

The OLED power connection in the current Wokwi setup uses the available ESP32 power connection.

12. CIRCUIT CONNECTIONS

OLED

OLED SDA → ESP32 GPIO 21

OLED SCL → ESP32 GPIO 22

OLED VCC → ESP32 power

OLED GND → ESP32 GND

Buzzer

Buzzer + → ESP32 GPIO 25

Buzzer - → ESP32 GND

Green LED

ESP32 GPIO 26

|

220Ω

|

Green LED

|

GND

Red LED

ESP32 GPIO 27

|

220Ω

|

Red LED

|

GND

13. SYSTEM ARCHITECTURE

The system is divided into five major sections:

13.1 Input Section

- The heartbeat-related information is represented by the pulse input concept.
- In the Wokwi implementation, BPM values are software-generated because the actual Pulse Sensor component is unavailable.

13.2 Processing Section

- The ESP32 processes the BPM value.

13.3 Decision Section

- The BPM value is compared with configured thresholds.

13.4 Output Section

- The OLED displays BPM and status.
- The LEDs and buzzer provide alerts.

13.5 Monitoring Section

- Serial Monitor provides debugging and real-time output information.
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14. BPM CONCEPT

BPM stands for Beats Per Minute.

It represents the number of heartbeats occurring in one minute.

The basic relationship between beat interval and BPM is:

$\text{BPM} = 60000 / \text{Time Between Two Beats in milliseconds}$

For example:

1000 ms → 60 BPM

750 ms → 80 BPM

600 ms → 100 BPM

In an actual sensor implementation, the microcontroller can detect successive heartbeat peaks and measure the time interval between them.

15. BPM SMOOTHING

Real sensor signals can contain noise.

Noise can result from:

- finger movement,
- sensor placement,
- electrical interference,
- environmental conditions,
- imperfect sensor contact.

Therefore, practical heartbeat-monitoring systems often use filtering, averaging or signal-processing techniques.

A moving average can help reduce sudden fluctuations.

The current Wokwi demonstration focuses primarily on the monitoring and alert architecture rather than advanced biomedical signal processing.

16. ALERT LOGIC

The project uses configurable educational thresholds.

`LOW_THRESHOLD = 60 BPM`

`HIGH_THRESHOLD = 100 BPM`

Normal Condition

$60 \leq \text{BPM} \leq 100$

Output:

Green LED = ON

Red LED = OFF

Buzzer = OFF

Status = NORMAL

Low Condition

$\text{BPM} < 60$

Output:

Green LED = OFF

Red LED = ON

Buzzer = ON

Status = LOW

High Condition

$\text{BPM} > 100$

Output:

Green LED = OFF

Red LED = ON

Buzzer = ON

Status = HIGH

These thresholds are configured for educational demonstration only and do not represent clinical diagnostic limits.

17. OLED DISPLAY

The OLED displays the current BPM and system status.

Normal

HEART MONITOR

BPM: 75

STATUS: NORMAL

Low

HEART MONITOR

BPM: 50

STATUS: LOW

High

HEART MONITOR

BPM: 110

STATUS: HIGH

18. ALERT SYSTEM

The alert system has two forms of feedback.

Visual Alert

- Red LED indicates an abnormal simulation condition.

Audible Alert

- The buzzer produces an alert during LOW or HIGH conditions.

Normal Indicator

The green LED indicates the normal configured range.

This provides a simple human-machine interface.

19. VIRTUAL SIMULATION

The project is simulated using Wokwi.

The simulation demonstrates the complete output and decision-making logic without requiring physical hardware.

The simulation cycles through different BPM conditions.

Example:

75 BPM

↓

NORMAL

↓

Green LED ON

↓

Buzzer OFF

50 BPM

↓

LOW

↓

Red LED ON

↓

Buzzer ON

110 BPM

↓

HIGH

↓

Red LED ON

↓

Buzzer ON

20. SIMULATION TESTING

The simulation is tested using three major BPM conditions.

Test 1: Normal BPM

Input:

75 BPM

Expected:

Status = NORMAL

Green LED = ON

Red LED = OFF

Buzzer = OFF

Test 2: Low BPM

Input:

50 BPM

Expected:

Status = LOW

Green LED = OFF

Red LED = ON

Buzzer = ON

Test 3: High BPM

Input:

110 BPM

Expected:

Status = HIGH

Green LED = OFF

Red LED = ON

Buzzer = ON

21. SERIAL MONITOR OUTPUT

The Serial Monitor provides debugging information.

Example:

Heartbeat Monitor Started

Heart Rate: 75 BPM | Status: NORMAL

Heart Rate: 75 BPM | Status: NORMAL

Heart Rate: 50 BPM | Status: LOW

Heart Rate: 50 BPM | Status: LOW

Heart Rate: 110 BPM | Status: HIGH

Heart Rate: 110 BPM | Status: HIGH

This output helps verify that the decision logic is working correctly.

22. TEST CASES

Test ID	Test	Input	Expected Status	Expected Alert
TC001	Normal BPM	75	NORMAL	No alert
TC002	Normal BPM	80	NORMAL	No alert
TC003	Low BPM	50	LOW	Red LED + Buzzer
TC004	Low BPM	55	LOW	Red LED + Buzzer
TC005	Low Threshold	59	LOW	Red LED + Buzzer
TC006	Lower Boundary	60	NORMAL	No alert
TC007	Normal BPM	70	NORMAL	No alert
TC008	Upper Boundary	100	NORMAL	No alert
TC009	High BPM	105	HIGH	Red LED + Buzzer
TC010	High BPM	110	HIGH	Red LED + Buzzer
TC011	OLED Test	75	NORMAL	Display verified
TC012	Buzzer Test	50	LOW	Buzzer verified
TC013	Red LED Test	110	HIGH	Red LED verified
TC014	Green LED Test	75	NORMAL	Green LED verified
TC015	Serial Test	75	NORMAL	Serial output verified
TC016	Alert Recovery	110 → 75	NORMAL	Alert stops

TC017 Full Simulation 75 → 50 → 110 NORMAL/LOW/HIGH All outputs verified

23. TESTING RESULT

The virtual simulation successfully demonstrates:

- Normal BPM processing
- Low BPM detection
- High BPM detection

- OLED status display
- Green LED activation
- Red LED activation
- Buzzer alert
- Serial Monitor output
- Alert recovery
- Automatic simulation cycle

Therefore, the embedded decision and alert logic works as expected in the educational simulation environment.

24. PROJECT FOLDER STRUCTURE

HEARTBEAT-MONITOR-ALERT-EMBEDDED-SYSTEM

GitHub Repository – Folder & File Structure

FOLDER / FILE STRUCTURE	Type	Description / Purpose
 HEARTBEAT-MONITOR-ALERT-EMBEDDED-SYSTEM/		
 arduino_code/	Folder	Contains Arduino source code
 heartbeat_monitor.ino	File (Arduino Sketch)	Main ESP32 code for heartbeat monitoring with alert system
 circuit_diagram/	Folder	Contains circuit diagram images
 01_complete-circuit.png	File (Image)	Complete circuit diagram of the project
 data/	Folder	Test data and results
 test-results.csv	File (CSV)	All test cases and expected vs actual results
 screenshots/	Folder	Project output screenshots
 01_complete-circuit.png	File (Image)	Complete circuit screenshot
 04_normal_bpm.png	File (Image)	Normal BPM output (Green LED ON)
 05_low_bpm_alert.png	File (Image)	Low BPM alert output (Red LED + Buzzer ON)
 06_high_bpm_alert.png	File (Image)	High BPM alert output (Red LED + Buzzer ON)
 07_serial_monitor.png	File (Image)	Serial Monitor output screenshot
 diag.png	File (Image)	Additional diagram or reference image
 simulation/	Folder	Wokwi simulation files
 diagram.json	File (Wokwi JSON)	Wokwi project configuration file
 reports/	Folder	Project documentation
 project-report.pdf	File (PDF)	Complete project report (PDF)
 outputs/	Folder	Sample outputs
 sample-output.txt	File (Text)	Sample serial monitor output (text format)
 README.md	File (Markdown)	Project overview, setup, usage, and details

Quick Summary:

➤ **arduino_code/** → Contains the main ESP32 code (.ino file)

➤ **circuit_diagram/** → Circuit diagram of the project

➤ **data/** → Test cases and results in CSV format

➤ **screenshots/** → All output and working screenshots

➤ **simulation/** → Wokwi simulation configuration

➤ **reports/** → Complete project report in PDF

➤ **outputs/** → Sample serial monitor outputs

➤ **README.md** → Project documentation and instructions

25. ADVANTAGES

The project provides several advantages as an educational embedded system:

1. Simple architecture.
2. Real-time decision-making demonstration.
3. Easy-to-understand alert mechanism.
4. OLED-based user interface.
5. Low-cost hardware concept.
6. ESP32 provides future connectivity possibilities.
7. Suitable for simulation.
8. Easy to expand with additional sensors.
9. Demonstrates embedded programming concepts.
10. Suitable as a GitHub proof-of-work project.

26. LIMITATIONS

The current project has the following limitations:

1. The Wokwi implementation does not use an actual Pulse Sensor.

2. BPM values are software-generated in the virtual simulation.
 3. The prototype is not medically certified.
 4. The configured thresholds are demonstration values.
 5. No clinical validation has been performed.
 6. Real-world sensor signals may contain noise.
 7. Sensor positioning can affect measurements.
 8. Advanced signal processing is not implemented in the basic version.
 9. The current simulation does not represent a complete medical monitoring system.
 10. The system should not be used for medical decisions.
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27. FUTURE ENHANCEMENTS

The project can be extended in several ways.

27.1 Real Pulse Sensor

- Integrate an actual Pulse Sensor with ESP32 and process the analog heartbeat waveform.

27.2 MAX30102

- Use the MAX30102 optical sensor for a more advanced sensor interface.

27.3 Advanced Signal Processing

Implement:

- noise filtering,
- peak detection,
- moving average,
- adaptive thresholding,
- signal validation.

27.4 Mobile Monitoring

ESP32 Wi-Fi can be used to send data to a mobile or web dashboard.

Possible technologies include:

- Blynk
- MQTT
- Web dashboard
- Cloud database

27.5 Data Logging

- Store BPM readings with timestamps for later analysis.

27.6 Battery Operation

- A suitable power-management system can make the prototype portable.

27.7 IoT Monitoring

- The system can be expanded into an IoT-based remote monitoring architecture.

28. INDUSTRY-ORIENTED EXTENSION

A more advanced architecture could be:

Heart-rate Sensor

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ESP32

|

v

Signal Processing

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v

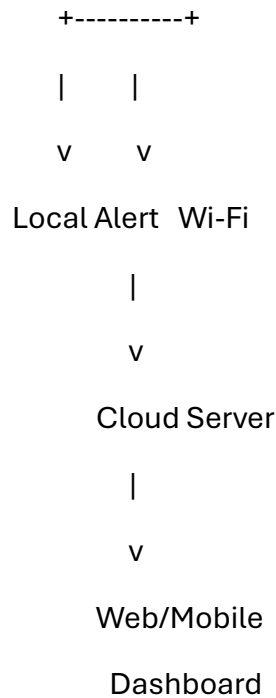
BPM Calculation

|

v

Threshold Engine

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This architecture demonstrates how a basic embedded prototype can evolve into an IoT monitoring system.

29. LEARNING OUTCOMES

After completing this project, the student gains experience in:

- ESP32 programming
- Embedded C/C++ programming
- GPIO configuration
- I2C communication
- OLED interfacing
- LED control
- Buzzer control
- Sensor interfacing concepts
- BPM calculation concepts
- Threshold-based decision making
- Real-time embedded logic
- Serial debugging

- Wokwi simulation
 - Test-case development
 - GitHub project management
 - Technical documentation
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30. GITHUB PROOF-OF-WORK

The project is documented as a GitHub repository.

Recommended repository name: Heartbeat-Monitor-Alert-Embedded-System

Recommended repository description:

- Embedded systems-based heartbeat monitoring prototype with BPM simulation, OLED display, and threshold-based buzzer and LED alerts using ESP32.

Recommended topics:

- embedded-systems
- arduino
- esp32
- heartbeat-monitor
- pulse-sensor
- biomedical
- embedded-c
- oled
- wokwi
- sensors

Important proof files include:

- Complete circuit screenshot
- Source code
- Normal BPM screenshot
- Low BPM alert screenshot
- High BPM alert screenshot
- Serial Monitor screenshot
- Testing CSV
- Project report

- README
 - Wokwi simulation files
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31. CONCLUSION

- The Heartbeat Monitor with Alert System demonstrates how an embedded microcontroller can be used to process heart-rate-related information and generate real-time visual and audible alerts.
 - The ESP32 acts as the central processing unit, while the OLED provides the user interface and the LEDs and buzzer provide status and alert feedback.
 - The Wokwi simulation successfully demonstrates normal, low and high BPM conditions. The project also demonstrates important embedded systems concepts including GPIO control, I2C communication, display interfacing, real-time processing, threshold comparison and system testing.
 - The project can be further enhanced by integrating a real pulse sensor, advanced signal processing, wireless communication, data logging and cloud-based monitoring.
 - This project should be considered an educational prototype only and not a medical diagnostic or clinical monitoring device.
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32. REFERENCES AND TOOLS

The following tools and technologies were used or considered for the project:

- ESP32 DevKit V1
 - Arduino-compatible programming environment
 - Wokwi simulation platform
 - OLED SSD1306 display
 - Adafruit GFX Library
 - Adafruit SSD1306 Library
 - GitHub for source-code management
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33. PROJECT STATUS

Current implementation:

Project Setup : Completed

ESP32 Configuration : Completed

OLED Interface : Completed

Buzzer Interface : Completed

Green LED Interface : Completed

Red LED Interface : Completed

Alert Logic : Completed

BPM Simulation : Completed

Wokwi Simulation : Completed

Testing Strategy : Completed

GitHub Documentation : completed

Real Pulse Sensor Version : Future Enhancement

Blynk Integration : Future Enhancement

34. FINAL PROJECT SUMMARY

The project can be summarized as:

INPUT

Heartbeat-related BPM information



PROCESSING

ESP32 Microcontroller



DECISION

Threshold Comparison



OUTPUT

OLED + Green LED + Red LED + Buzzer



RESULT

Real-time status and alert demonstration

The project provides a practical demonstration of embedded-system design and can serve as a foundation for more advanced wearable, IoT and biomedical monitoring prototypes.

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Name: Sonamika Anand Samrat

Project: Heartbeat Monitor with Alert System

Domain: Embedded Systems

Platform: ESP32 + Wokwi

Project Type: Educational Embedded Systems Prototype